RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2018- 2019 (Odd Semester)

<u>Innovative Practices Description</u>

UNIT V WAVE GUIDES AND CAVITY RESONATORS

Degree, Semester& Branch: V Semester B.E. ECE B
Course Code & Title: EC6503 Transmission Lines and Waveguides
Name of the Faculty member: Mr.D.Gopinath
Name of the Topic: TM wave in Rectangular wave guides
Name of the Innovative Practice: Video Lecture
ICT Tool Used:- Smart Class Room

<u>Description of Video Lecture:</u>

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Rectangular waveguides are the one of the earliest type of the transmission lines. They are used in many applications. A lot of components such as isolators, detectors, attenuators, couplers and slotted lines are available for various standard waveguide bands between 1 GHz to above 220 GHz.

A rectangular waveguide supports TM and TE modes but not TEM waves because we cannot define a unique voltage since there is only one conductor in a rectangular waveguide. The shape of a rectangular waveguide is as shown below. A material with permittivity e and permeability m fills the inside of the conductor.

I have shown NPTEL Video Lecture series on "Transmission Lines and E.M Waves" by Prof. R.K.Shevgaonkar, Department of Electrical Engineering, IIT Bombay. In that Lecture No.38 about Rectangular waveguides. After shown this video, students can be identified the difference between Transverse Electric (TE) and Transverse Magnetic (TM) waves propagated through rectangular waveguides.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	2	2	1					1	1		

CO5: The student will be able to apply the fundamental concept of transmission in different guided systems and Utilize cavity resonators.

References:

1. John D Ryder, “Networks, lines and fields”, 2nd Edition, Prentice Hall India, 2010.
2. G.S.N Raju "Electromagnetic Field Theory and Transmission Lines Pearson Education, First edition 2005.
3. https://www.youtube.com/watch?v=6ZdXP4ABtaQ&list=PL0CD49F1FAD4E6FAA&index=38

Signature of the Faculty Member **HOD/ECE**